

MAT 426, Class 4 Homework: LU Factorization with Partial Pivoting

(Due Thursday, February 19, 2026)

In Exercises 1–4, find the $PA = LU$ factorization of the matrix A .

1. $A = \begin{bmatrix} 1 & 2 & 4 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

2. $A = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 5 & 6 \\ 1 & 2 & 4 \end{bmatrix}$

3. $A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -7 & 3 \\ 1 & 1 & 3 \end{bmatrix}$

4. $A = \begin{bmatrix} 2 & -6 & 6 \\ -4 & 5 & -7 \\ 3 & 5 & -1 \\ -6 & 4 & -8 \end{bmatrix}$

In Exercises 5–6, consider a 4×4 matrix A . Using Gaussian elimination with partial pivoting yields $M_3P_3M_2P_2M_1P_1A = U$.

5. Compute $P_2P_3P_3P_2$

6. Use the result from Exercise 5 to write $M'_3M'_2M'_1PA = U$. Clearly identify how the M'_k and P are determined. Note: This will require two uses of the “multiply by one” tool.